R Data Types and Transformations with dplyr

PSYC 259: Principles of Data Science

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How does R data processing help to minimize mistakes?

How can R data processing help us minimize mistakes?

- Code *is* documentation
- Handle data with less human intervention (e.g., `read_tsv()` in the homework)
- But coding errors are also common...what can we do to make sure that our code works?
  - Write reproducible code (avoiding absolute file paths)
  - Use code to clean up variable names to make them human-readable
  - Instead of "hard" coding based on position (such as ds[1, 2]), filter and select by name and logical conditions
  - Reduce code duplication (copy-paste-tweak) with loops and functions

Data type and transformation tutorial

- Part 1:
  - Data types and classes in R
  - Logical statements
- Part 2: Introduction to *dplyr*
- Part 3: Code style

Follow along from the Github repo

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Part 1: Data Types and Classes; Logical Comparisons

Data types and classes

- Allows language to efficiently store data in memory (*type* in R)
    - If a value is an integer (1, 2, 3, 4) storing it as an integer makes calculations easier compared to storing it as a double (2.34542480424624086)
    - integer, double, logical, and character are R types
- Allows language to follow transformation rules (*class* in R)
    - Addition/subtraction for a date follows different rules compared with integers/double
    - “Less than” makes sense when comparing numbers, but not when comparing strings
    - The class of an object tells the language how to handle it

Common data types/classes in R

- Numeric class
    - integer type: 1, 2, 3
    - double type: 1.12124
- Character type/class: “hello”
- Logical type/class: T/F (TRUE/FALSE)
- Datetime class (type is double)
- Factor class (type is integer)

Functions to test and convert types

- Inspect the type/class
  - `typeof()`, `class()`, and `str()`
  - `dplyr::glimpse()` for data frames
- Test if something is a particular class
  - `is.numeric()`, `is.character()`
  - returns a logical (T/F)
- Take a value and convert it into a different class
  - `as.numeric(x)`, `as.character(x)`, `as.date(x)`
  - returns a typed/classed version of x

Data types

```
x <- 1
```

Data types

```
x <- 1
typeof(x)
```

```
[1] "double"
```

Data types

```
x <- 1
typeof(x)
class(x)
```

```
[1] "double"
```

```
[1] "numeric"
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
is.numeric(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA
```

Data types

```
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
try(x + ", World")
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA

Error in x + ", World" : non-numeric argu
```

Data types

```r
x <- 1
typeof(x)
class(x)
is.numeric(x)
is.character(x)
as.character(x)
x + 1

x <- "Hello"
typeof(x)
is.numeric(x)
is.character(x)
as.numeric(x)
try(x + ", World")
paste0(x, ", World")
```

```
[1] "double"

[1] "numeric"

[1] TRUE

[1] FALSE

[1] "1"

[1] 2

[1] "character"

[1] FALSE

[1] TRUE

[1] NA

Error in x + ", World" : non-numeric argu

[1] "Hello, World"
```

R tries to coerce classes/types to do what you ask

- `+` is a function in `R` (you can actually call it `` `+`(1, 2) ``)
- `"Hello" + ", World"` fails because the `+` function requires numeric arguments
- paste0 is happy to take numeric arguments and will coerce them to character
- `paste0(1, 2)` returns `"12"`
- What happens if we evaluate `as.numeric(paste0(1,2))`?
- What happens if we evaluate `TRUE + TRUE`?

Built-in logical comparison functions

- Comparisons between values that result in TRUE/FALSE

Function	Operator	Example
Numerical comparison	`>`, `>=`, `<`, `<=`	`0 < 1`
Equals	`==`	`1 == 1`
Not equals	`!=`	`1 != 2`
And	`&`	`1 > 0 & 0 > 1`
Or	|	1 > 0 | 0 > 1
Not	`!`	`!(1 > 0 & 0 > 1)`
Matching	`%in%`	`1 %in% c(-1, 0, 1)`

How do we use logical statements?

- Subsetting logic (filtering, selecting)
- Through the `ifelse()` command
  - `ifelse(LOGICAL STATEMENT, DO IF TRUE, DO IF FALSE)`

```
x <- c(0, 1, 2, 3, NA)
ifelse(NA %in% x, "x contains a missing value", "x does not contain a missing value")
```

```
[1] "x contains a missing value"
```

```
x <- c(0, 1, 2, 3)
ifelse(NA %in% x, "x contains a missing value", "x does not contain a missing value")
```

```
[1] "x does not contain a missing value"
```

Part 2: Data transformation with the dplyr package

Data transformation with the dplyr package

- A toolbox for common data processing/manipulation operations to apply to tibbles
  - `glimpse()` to see the structure of a tibble
  - `arrange()` to sort data by columns
  - `filter()`, `filter_out()` and `slice()` to subset data by rows
  - `select()` to subset data by columns
  - `rename()` and `relocate()` to rename and move columns
  - `mutate()` to add or change columns (or their values)
  - `summarize()` to calculate summary statistics over rows of data
  - `group_by()` to perform operations within subsets of data
  - and many more...

Each dplyr function uses a similar structure

- `function(data, something_to_do_with_columns)`
- In base R, you need to specify the dataset over and over to subset: `ds[ds$col1 == 1, ]`
- In dplyr functions, the data argument lets you work with column names without using `$`. The data argument also lets you access the columns directly without using quoted expressions: `filter(data, col1 == 1)`
- Most dplyr functions return the entire tibble back as an output
- Read help pages (e.g., `?filter`) to understand what arguments each function accepts and what each function returns

Arrange (sorting data)

```
library(tidyverse)  #loads dplyr
ds <- starwars  #loads built-in star wars database
glimpse(ds)
```

```
Rows: 87
Columns: 14
$ name       <chr> "Luke Skywalker", "C-3PO", "R2-D2", "Darth Vader", "Leia Or…
$ height     <int> 172, 167, 96, 202, 150, 178, 165, 97, 183, 182, 188, 180, 2…
$ mass       <dbl> 77.0, 75.0, 32.0, 136.0, 49.0, 120.0, 75.0, 32.0, 84.0, 77.…
$ hair_color <chr> "blond", NA, NA, "none", "brown", "brown, grey", "brown", N…
$ skin_color <chr> "fair", "gold", "white, blue", "white", "light", "light", "…
$ eye_color  <chr> "blue", "yellow", "red", "yellow", "brown", "blue", "blue",…
$ birth_year <dbl> 19.0, 112.0, 33.0, 41.9, 19.0, 52.0, 47.0, NA, 24.0, 57.0, …
$ sex        <chr> "male", "none", "none", "male", "female", "male", "female",…
$ gender     <chr> "masculine", "masculine", "masculine", "masculine", "femini…
$ homeworld  <chr> "Tatooine", "Tatooine", "Naboo", "Tatooine", "Alderaan", "T…
$ species    <chr> "Human", "Droid", "Droid", "Human", "Human", "Human", "Huma…
$ films      <list> <"A New Hope", "The Empire Strikes Back", "Return of the J…
$ vehicles   <list> <"Snowspeeder", "Imperial Speeder Bike">, <>, <>, <>, "Imp…
$ starships  <list> <"X-wing", "Imperial shuttle">, <>, <>, "TIE Advanced x1",…
```

Arrange (sorting data)

```r
library(tidyverse)  #loads dplyr
ds <- starwars  #loads built-in star wars database
arrange(ds, name)
```

```
# A tibble: 87 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 Ackbar      180    83 none       brown mot… orange           41   male  mascu…
 2 Adi Gal…    184    50 none       dark       blue             NA   fema… femin…
 3 Anakin …    188    84 blond      fair       blue             41.9 male  mascu…
 4 Arvel C…     NA    NA brown      fair       brown            NA   male  mascu…
 5 Ayla Se…    178    55 none       blue       hazel            48   fema… femin…
 6 BB8          NA    NA none       none       black            NA   none  mascu…
 7 Bail Pr…    191    NA black      tan        brown            67   male  mascu…
 8 Barriss…    166    50 black      yellow     blue             40   fema… femin…
 9 Ben Qua…    163    65 none       grey, gre… orange           NA   male  mascu…
10 Beru Wh…    165    75 brown      light      blue             47   fema… femin…
# ℹ 77 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Arrange (sorting data)

```r
library(tidyverse)  #loads dplyr
ds <- starwars  #loads built-in star wars database
arrange(ds, desc(height), mass)
```

```
# A tibble: 87 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 Yarael …    264    NA none       white      yellow           NA   male  mascu…
 2 Tarfful     234   136 brown      brown      blue             NA   male  mascu…
 3 Lama Su     229    88 none       grey       black            NA   male  mascu…
 4 Chewbac…    228   112 brown      unknown    blue            200   male  mascu…
 5 Roos Ta…    224    82 none       grey       orange           NA   male  mascu…
 6 Grievous    216   159 none       brown, wh… green, y…        NA   male  mascu…
 7 Taun We     213    NA none       grey       black            NA   fema… femin…
 8 Tion Me…    206    80 none       grey       black            NA   male  mascu…
 9 Rugor N…    206    NA none       green      orange           NA   male  mascu…
10 Darth V…    202   136 none       white      yellow           41.9 male  mascu…
# ℹ 77 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Arrange (sorting data)

```
library(tidyverse)  #loads dplyr
ds <- starwars  #loads built-in star wars database
arrange(ds, eye_color, hair_color)
```

```
# A tibble: 87 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 Nien Nu…    160    68 none       grey       black              NA male  mascu…
 2 Gasgano     122    NA none       white, bl… black              NA male  mascu…
 3 Kit Fis…    196    87 none       green      black              NA male  mascu…
 4 Plo Koon    188    80 none       orange     black              22 male  mascu…
 5 Lama Su     229    88 none       grey       black              NA male  mascu…
 6 Taun We     213    NA none       grey       black              NA fema… femin…
 7 Shaak Ti    178    57 none       red, blue… black              NA fema… femin…
 8 Tion Me…    206    80 none       grey       black              NA male  mascu…
 9 BB8          NA    NA none       none       black              NA none  mascu…
10 Greedo      173    74 <NA>       green      black              44 male  mascu…
# ℹ 77 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

What are some uses of arrange()?

Filter (subsetting rows)

```
filter(ds, name == "Yoda")
```

```
# A tibble: 1 × 14
  name  height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>  <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Yoda      66    17 white      green      brown             896 male  masculine
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, is.na(hair_color))
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 C-3PO        167    75 <NA>       gold       yellow            112 none  mascu…
2 R2-D2         96    32 <NA>       white, bl… red                33 none  mascu…
3 R5-D4         97    32 <NA>       white, red red                NA none  mascu…
4 Greedo       173    74 <NA>       green      black              44 male  mascu…
5 Jabba De…    175  1358 <NA>       green-tan… orange            600 herm… mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, height > 100, height < 150)
```

```
# A tibble: 3 × 14
  name    height  mass hair_color skin_color  eye_color birth_year sex   gender
  <chr>    <int> <dbl> <chr>      <chr>       <chr>           <dbl> <chr> <chr>
1 Watto      137    NA black      blue, grey  yellow             NA male  mascul…
2 Sebulba    112    40 none       grey, red   orange             NA male  mascul…
3 Gasgano    122    NA none       white, blue black              NA male  mascul…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, eye_color %in% c("blue","brown"))
```

```
# A tibble: 40 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 Luke Sk…    172    77 blond      fair       blue             19   male  mascu…
 2 Leia Or…    150    49 brown      light      brown            19   fema… femin…
 3 Owen La…    178   120 brown, gr… light      blue             52   male  mascu…
 4 Beru Wh…    165    75 brown      light      blue             47   fema… femin…
 5 Biggs D…    183    84 black      light      brown            24   male  mascu…
 6 Anakin …    188    84 blond      fair       blue             41.9 male  mascu…
 7 Wilhuff…    180    NA auburn, g… fair       blue             64   male  mascu…
 8 Chewbac…    228   112 brown      unknown    blue            200   male  mascu…
 9 Han Solo    180    80 brown      fair       brown            29   male  mascu…
10 Jek Ton…    180   110 brown      fair       blue             NA   <NA>  <NA>
# ℹ 30 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, !(eye_color %in% c("blue","brown")))
```

```
# A tibble: 47 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 C-3PO       167    75 <NA>       gold       yellow          112   none  mascu…
 2 R2-D2        96    32 <NA>       white, bl… red              33   none  mascu…
 3 Darth V…    202   136 none       white      yellow           41.9 male  mascu…
 4 R5-D4        97    32 <NA>       white, red red              NA   none  mascu…
 5 Obi-Wan…    182    77 auburn, w… fair       blue-gray        57   male  mascu…
 6 Greedo      173    74 <NA>       green      black            44   male  mascu…
 7 Jabba D…    175  1358 <NA>       green-tan… orange          600   herm… mascu…
 8 Wedge A…    170    77 brown      fair       hazel            21   male  mascu…
 9 Palpati…    170    75 grey       pale       yellow           82   male  mascu…
10 IG-88       200   140 none       metal      red              15   none  mascu…
# ℹ 37 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter(ds, height < 100)
```

```
# A tibble: 7 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 R2-D2         96    32 <NA>       white, bl… red                33 none  mascu…
2 R5-D4         97    32 <NA>       white, red red                NA none  mascu…
3 Yoda          66    17 white      green      brown             896 male  mascu…
4 Wicket S…     88    20 brown      brown      brown               8 male  mascu…
5 Ratts Ty…     79    15 none       grey, blue unknown            NA male  mascu…
6 Dud Bolt      94    45 none       blue, grey yellow             NA male  mascu…
7 R4-P17        96    NA none       silver, r… red, blue          NA none  femin…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Filter (subsetting rows)

```
filter_out(ds, height >= 100)
```

```
# A tibble: 13 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 R2-D2        96    32 <NA>       white, bl… red                33 none  mascu…
 2 R5-D4        97    32 <NA>       white, red red                NA none  mascu…
 3 Yoda         66    17 white      green      brown             896 male  mascu…
 4 Arvel C…     NA    NA brown      fair       brown              NA male  mascu…
 5 Wicket …     88    20 brown      brown      brown               8 male  mascu…
 6 Ratts T…     79    15 none       grey, blue unknown            NA male  mascu…
 7 Dud Bolt     94    45 none       blue, grey yellow             NA male  mascu…
 8 R4-P17       96    NA none       silver, r… red, blue          NA none  femin…
 9 Finn         NA    NA black      dark       dark               NA male  mascu…
10 Rey          NA    NA brown      light      hazel              NA fema… femin…
11 Poe Dam…     NA    NA brown      light      brown              NA male  mascu…
12 BB8          NA    NA none       none       black              NA none  mascu…
13 Captain…     NA    NA none       none       unknown            NA fema… femin…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Why do `filter(height < 100)` and `filter_out(height >= 100)` return different results?

Why not subset based on position?

```
filter(starwars, name == "Darth Vader")
```

```
# A tibble: 1 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Darth Va…    202   136 none       white      yellow           41.9 male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

```
starwars[4,]
```

```
# A tibble: 1 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Darth Va…    202   136 none       white      yellow           41.9 male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice(ds, 10:15)
```

```
# A tibble: 6 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Obi-Wan …    182    77 auburn, w… fair       blue-gray        57   male  mascu…
2 Anakin S…    188    84 blond      fair       blue             41.9 male  mascu…
3 Wilhuff …    180    NA auburn, g… fair       blue             64   male  mascu…
4 Chewbacca    228   112 brown      unknown    blue            200   male  mascu…
5 Han Solo     180    80 brown      fair       brown            29   male  mascu…
6 Greedo       173    74 <NA>       green      black            44   male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice_head(ds, n = 5)
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Luke Sky…    172    77 blond      fair       blue             19   male  mascu…
2 C-3PO        167    75 <NA>       gold       yellow          112   none  mascu…
3 R2-D2         96    32 <NA>       white, bl… red              33   none  mascu…
4 Darth Va…    202   136 none       white      yellow           41.9 male  mascu…
5 Leia Org…    150    49 brown      light      brown            19   fema… femin…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice_tail(ds, n = 5)
```

```
# A tibble: 5 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Finn          NA    NA black      dark       dark               NA male  mascu…
2 Rey           NA    NA brown      light      hazel              NA fema… femin…
3 Poe Dame…     NA    NA brown      light      brown              NA male  mascu…
4 BB8           NA    NA none       none       black              NA none  mascu…
5 Captain …     NA    NA none       none       unknown            NA fema… femin…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice_sample(ds, n = 2)
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Palpatine    170    75 grey       pale       yellow             82 male  mascu…
2 Mace Win…    188    84 none       dark       brown              72 male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice_sample(ds, n = 2)
```

```
# A tibble: 2 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Leia Org…    150    49 brown      light      brown              19 fema… femin…
2 Han Solo     180    80 brown      fair       brown              29 male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Slice (other kinds of subsets)

```
slice_min(ds, height, n = 3)
```

```
# A tibble: 3 × 14
  name      height  mass hair_color skin_color eye_color birth_year sex   gender
  <chr>      <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
1 Yoda          66    17 white      green      brown             896 male  mascu…
2 Ratts Ty…     79    15 none       grey, blue unknown            NA male  mascu…
3 Wicket S…     88    20 brown      brown      brown              8 male  mascu…
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Select (subsetting columns)

```
select(ds, name, height, mass)
```

```
# A tibble: 87 × 3
   name               height  mass
   <chr>               <int> <dbl>
 1 Luke Skywalker        172    77
 2 C-3PO                 167    75
 3 R2-D2                  96    32
 4 Darth Vader           202   136
 5 Leia Organa           150    49
 6 Owen Lars             178   120
 7 Beru Whitesun Lars    165    75
 8 R5-D4                  97    32
 9 Biggs Darklighter     183    84
10 Obi-Wan Kenobi        182    77
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, c("name", "height", "mass"))
```

```
# A tibble: 87 × 3
   name               height  mass
   <chr>               <int> <dbl>
 1 Luke Skywalker        172    77
 2 C-3PO                 167    75
 3 R2-D2                  96    32
 4 Darth Vader           202   136
 5 Leia Organa           150    49
 6 Owen Lars             178   120
 7 Beru Whitesun Lars    165    75
 8 R5-D4                  97    32
 9 Biggs Darklighter     183    84
10 Obi-Wan Kenobi        182    77
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, name:eye_color)
```

```
# A tibble: 87 × 6
   name               height  mass hair_color    skin_color  eye_color
   <chr>               <int> <dbl> <chr>         <chr>       <chr>
 1 Luke Skywalker        172    77 blond         fair        blue
 2 C-3PO                 167    75 <NA>          gold        yellow
 3 R2-D2                  96    32 <NA>          white, blue red
 4 Darth Vader           202   136 none          white       yellow
 5 Leia Organa           150    49 brown         light       brown
 6 Owen Lars             178   120 brown, grey   light       blue
 7 Beru Whitesun Lars    165    75 brown         light       blue
 8 R5-D4                  97    32 <NA>          white, red  red
 9 Biggs Darklighter     183    84 black         light       brown
10 Obi-Wan Kenobi        182    77 auburn, white fair        blue-gray
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, -(eye_color:starships))
```

```
# A tibble: 87 × 5
   name               height  mass hair_color    skin_color
   <chr>               <int> <dbl> <chr>         <chr>
 1 Luke Skywalker        172    77 blond         fair
 2 C-3PO                 167    75 <NA>          gold
 3 R2-D2                  96    32 <NA>          white, blue
 4 Darth Vader           202   136 none          white
 5 Leia Organa           150    49 brown         light
 6 Owen Lars             178   120 brown, grey   light
 7 Beru Whitesun Lars    165    75 brown         light
 8 R5-D4                  97    32 <NA>          white, red
 9 Biggs Darklighter     183    84 black         light
10 Obi-Wan Kenobi        182    77 auburn, white fair
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, ends_with("color"))
```

```
# A tibble: 87 × 3
   hair_color    skin_color  eye_color
   <chr>         <chr>       <chr>
 1 blond         fair        blue
 2 <NA>          gold        yellow
 3 <NA>          white, blue red
 4 none          white       yellow
 5 brown         light       brown
 6 brown, grey   light       blue
 7 brown         light       blue
 8 <NA>          white, red  red
 9 black         light       brown
10 auburn, white fair        blue-gray
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, contains("_"))
```

```
# A tibble: 87 × 4
   hair_color    skin_color  eye_color birth_year
   <chr>         <chr>       <chr>           <dbl>
 1 blond         fair        blue             19
 2 <NA>          gold        yellow          112
 3 <NA>          white, blue red              33
 4 none          white       yellow           41.9
 5 brown         light       brown            19
 6 brown, grey   light       blue             52
 7 brown         light       blue             47
 8 <NA>          white, red  red              NA
 9 black         light       brown            24
10 auburn, white fair        blue-gray        57
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, where(is.numeric))
```

```
# A tibble: 87 × 3
   height  mass birth_year
    <int> <dbl>      <dbl>
 1    172    77       19
 2    167    75      112
 3     96    32       33
 4    202   136       41.9
 5    150    49       19
 6    178   120       52
 7    165    75       47
 8     97    32       NA
 9    183    84       24
10    182    77       57
# ℹ 77 more rows
```

Select (subsetting columns)

```
select(ds, where(is.character))
```

```
# A tibble: 87 × 8
   name           hair_color skin_color eye_color sex   gender homeworld species
   <chr>          <chr>      <chr>      <chr>     <chr> <chr>  <chr>     <chr>
 1 Luke Skywalker blond      fair       blue      male  mascu… Tatooine  Human
 2 C-3PO          <NA>       gold       yellow    none  mascu… Tatooine  Droid
 3 R2-D2          <NA>       white, bl… red       none  mascu… Naboo     Droid
 4 Darth Vader    none       white      yellow    male  mascu… Tatooine  Human
 5 Leia Organa    brown      light      brown     fema… femin… Alderaan  Human
 6 Owen Lars      brown, gr… light      blue      male  mascu… Tatooine  Human
 7 Beru Whitesun… brown      light      blue      fema… femin… Tatooine  Human
 8 R5-D4          <NA>       white, red red       none  mascu… Tatooine  Droid
 9 Biggs Darklig… black      light      brown     male  mascu… Tatooine  Human
10 Obi-Wan Kenobi auburn, w… fair       blue-gray male  mascu… Stewjon   Human
# ℹ 77 more rows
```

Select needs good column names

```
bad_names <- tibble(id = 1, depression_1 = 1:5, dep2 = 1:5, DEP3 = 1:5, depressed = T)
select(bad_names, starts_with("dep"))
```

```
# A tibble: 5 × 4
  depression_1  dep2  DEP3 depressed
         <int> <int> <int> <lgl>
1            1     1     1 TRUE
2            2     2     2 TRUE
3            3     3     3 TRUE
4            4     4     4 TRUE
5            5     5     5 TRUE
```

Select needs good column names

```
bad_names <- tibble(id = 1, depression_1 = 1:5, dep2 = 1:5, DEP3 = 1:5, depressed = T)
select(bad_names, starts_with("dep") & !matches("depressed"))
```

```
# A tibble: 5 × 3
  depression_1  dep2  DEP3
         <int> <int> <int>
1            1     1     1
2            2     2     2
3            3     3     3
4            4     4     4
5            5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)
select(good_names, starts_with("dep"))
```

```
# A tibble: 5 × 3
  dep_1 dep_2 dep_3
  <int> <int> <int>
1     1     1     1
2     2     2     2
3     3     3     3
4     4     4     4
5     5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)
select(good_names, matches("_[0-9]"))
```

```
# A tibble: 5 × 3
  dep_1 dep_2 dep_3
  <int> <int> <int>
1     1     1     1
2     2     2     2
3     3     3     3
4     4     4     4
5     5     5     5
```

Select needs good column names

```
good_names <- tibble(id = 1, dep_1 = 1:5, dep_2 = 1:5, dep_3 = 1:5, is_depressed = T)
select(good_names, num_range(prefix = "dep_", range = 1:3))
```

```
# A tibble: 5 × 3
  dep_1 dep_2 dep_3
  <int> <int> <int>
1     1     1     1
2     2     2     2
3     3     3     3
4     4     4     4
5     5     5     5
```

What's going on here?

```
select(ds, name, height, eye_color)
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows
```

What's going on here?

```
select(ds, name, height, eye_color)
filter(ds, height < 70)
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows

# A tibble: 1 × 14
  name  height  mass hair_color skin_color eye_color birth_
  <chr>  <int> <dbl> <chr>      <chr>      <chr>           <
1 Yoda      66    17 white      green      brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>
```

What's going on here?

```
select(ds, name, height, eye_color)
filter(ds, height < 70)
ds
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows

# A tibble: 1 × 14
  name  height  mass hair_color skin_color eye_color birth_
  <chr>  <int> <dbl> <chr>      <chr>      <chr>          <
1 Yoda      66    17 white      green      brown
# ℹ 5 more variables: homeworld <chr>, species <chr>, films
#   vehicles <list>, starships <list>

# A tibble: 87 × 14
   name     height  mass hair_color skin_color eye_color bi
   <chr>     <int> <dbl> <chr>      <chr>      <chr>
 1 Luke Sk…    172    77 blond      fair       blue
 2 C-3PO       167    75 <NA>       gold       yellow
 3 R2-D2        96    32 <NA>       white, bl… red
 4 Darth V…    202   136 none       white      yellow
 5 Leia Or…    150    49 brown      light      brown
 6 Owen La…    178   120 brown, gr… light      blue
```

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)

ds
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows
```

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)
ds <- arrange(ds, height, eye_color)

ds
```

```
# A tibble: 87 × 3
   name                  height eye_color
   <chr>                  <int> <chr>
 1 Yoda                      66 brown
 2 Ratts Tyerel              79 unknown
 3 Wicket Systri Warrick     88 brown
 4 Dud Bolt                  94 yellow
 5 R2-D2                     96 red
 6 R4-P17                    96 red, blue
 7 R5-D4                     97 red
 8 Sebulba                  112 orange
 9 Gasgano                  122 black
10 Watto                    137 yellow
# ℹ 77 more rows
```

Reassign the transformations back to the tibble

```
ds <- select(ds, name, height, eye_color)
ds <- arrange(ds, height, eye_color)
ds <- filter(ds, height < 70)

ds
```

```
# A tibble: 1 × 3
  name  height eye_color
  <chr>  <int> <chr>
1 Yoda       66 brown
```

Why isn't this a good strategy?

```
ds <- starwars
```

Why isn't this a good strategy?

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
```

Why isn't this a good strategy?

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
```

Why isn't this a good strategy?

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
ds_sorted_filtered <- filter(ds_sorted, height < 70)
```

Why isn't this a good strategy?

```
ds <- starwars
ds_name_height_eye_color <- select(ds, name, height, eye_color)
ds_sorted <- arrange(ds_name_height_eye_color, height, eye_color)
ds_sorted_filtered <- filter(ds_sorted, height < 70)
ds_sorted_filtered
```

```
# A tibble: 1 × 3
  name  height eye_color
  <chr>  <int> <chr>
1 Yoda      66 brown
```

Introducing the pipe operator: %>%

```
ds <- starwars

ds
```

```
# A tibble: 87 × 14
   name     height  mass hair_color skin_color eye_color birth_year sex   gender
   <chr>     <int> <dbl> <chr>      <chr>      <chr>           <dbl> <chr> <chr>
 1 Luke Sk…    172    77 blond      fair       blue             19   male  mascu…
 2 C-3PO       167    75 <NA>       gold       yellow          112   none  mascu…
 3 R2-D2        96    32 <NA>       white, bl… red              33   none  mascu…
 4 Darth V…    202   136 none       white      yellow           41.9 male  mascu…
 5 Leia Or…    150    49 brown      light      brown            19   fema… femin…
 6 Owen La…    178   120 brown, gr… light      blue             52   male  mascu…
 7 Beru Wh…    165    75 brown      light      blue             47   fema… femin…
 8 R5-D4        97    32 <NA>       white, red red              NA   none  mascu…
 9 Biggs D…    183    84 black      light      brown            24   male  mascu…
10 Obi-Wan…    182    77 auburn, w… fair       blue-gray        57   male  mascu…
# ℹ 77 more rows
# ℹ 5 more variables: homeworld <chr>, species <chr>, films <list>,
#   vehicles <list>, starships <list>
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds

ds
```

```
# A tibble: 87 × 3
   name               height eye_color
   <chr>               <int> <chr>
 1 Luke Skywalker        172 blue
 2 C-3PO                 167 yellow
 3 R2-D2                  96 red
 4 Darth Vader           202 yellow
 5 Leia Organa           150 brown
 6 Owen Lars             178 blue
 7 Beru Whitesun Lars    165 blue
 8 R5-D4                  97 red
 9 Biggs Darklighter     183 brown
10 Obi-Wan Kenobi        182 blue-gray
# ℹ 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds %>%
  select(name, eye_color)

ds
```

```
# A tibble: 87 × 2
   name               eye_color
   <chr>              <chr>
 1 Luke Skywalker     blue
 2 C-3PO              yellow
 3 R2-D2              red
 4 Darth Vader        yellow
 5 Leia Organa        brown
 6 Owen Lars          blue
 7 Beru Whitesun Lars blue
 8 R5-D4              red
 9 Biggs Darklighter  brown
10 Obi-Wan Kenobi     blue-gray
# ℹ 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds %>%
  select(name, eye_color) %>%
  arrange(eye_color)

ds
```

```
# A tibble: 87 × 2
   name       eye_color
   <chr>      <chr>
 1 Greedo     black
 2 Nien Nunb  black
 3 Gasgano    black
 4 Kit Fisto  black
 5 Plo Koon   black
 6 Lama Su    black
 7 Taun We    black
 8 Shaak Ti   black
 9 Tion Medon black
10 BB8        black
# ℹ 77 more rows
```

Introducing the pipe operator: %>%

```
ds <- starwars
ds <- ds %>% select(name, height, eye_color)

ds <- ds %>%
  select(name, eye_color) %>%
  arrange(eye_color) %>%
  filter(eye_color == "blue")

ds
```

```
# A tibble: 19 × 2
   name               eye_color
   <chr>              <chr>
 1 Luke Skywalker     blue
 2 Owen Lars          blue
 3 Beru Whitesun Lars blue
 4 Anakin Skywalker   blue
 5 Wilhuff Tarkin     blue
 6 Chewbacca          blue
 7 Jek Tono Porkins   blue
 8 Lobot              blue
 9 Mon Mothma         blue
10 Qui-Gon Jinn       blue
11 Finis Valorum      blue
12 Ric Olié           blue
13 Adi Gallia         blue
14 Mas Amedda         blue
15 Cliegg Lars        blue
16 Luminara Unduli    blue
17 Barriss Offee      blue
18 Jocasta Nu         blue
19 Tarfful            blue
```

Assigning the result of a piped expression

All the options below are equivalent, but I stick with `ds <- ds %>%` because it's the most common

```
ds <- ds %>% select(height) %>% slice_tail(n = 5)

ds %>% select(height) %>% slice_tail(n = 5) -> ds

ds <- select(ds, height) %>% slice_tail(n = 5)

ds <- slice_tail(select(ds, height), n = 5)
```

Learn your keyboard shortcuts to save time

Rename

- Using dplyr's `rename` and `rename_with` can help clean up messy names from other sources
    - Avoid names that contain spaces or start with numbers
    - Force names to use a similar format like snake_case or camelCase
- Installing and loading the `janitor` package opens up some helpful renaming tools

Rename columns

```
# Switching to built-in iris data set since starwars has good names
iris
```

```
    Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns

```
# Switching to built-in iris data set since starwars has good names
iris %>% rename(sepal_length = Sepal.Length)
```

```
    sepal_length Sepal.Width Petal.Length Petal.Width    Species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns

```
# Switching to built-in iris data set since starwars has good names
iris %>% rename(sepal_length = Sepal.Length, sepal_width = Sepal.Width, petal_length = Petal
```

```
    sepal_length sepal_width petal_length petal_width    species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns

```
# Switching to built-in iris data set since starwars has good names
iris %>% rename_with(toupper)
```

```
    SEPAL.LENGTH SEPAL.WIDTH PETAL.LENGTH PETAL.WIDTH    SPECIES
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns

```
# Switching to built-in iris data set since starwars has good names
iris %>% rename_with(tolower, starts_with("Petal"))
```

```
    Sepal.Length Sepal.Width petal.length petal.width    Species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns with a lookup table

```
lookup <- c(pet_length = "Petal.Length", pet_width = "Petal.Width",
            sep_length = "Sepal.Length", sep_width = "Sepal.Width")
```

Rename columns with a lookup table

```
lookup <- c(pet_length = "Petal.Length", pet_width = "Petal.Width",
            sep_length = "Sepal.Length", sep_width = "Sepal.Width")
iris %>% rename(all_of(lookup))
```

```
    sep_length sep_width pet_length pet_width    Species
1          5.1       3.5        1.4       0.2     setosa
2          4.9       3.0        1.4       0.2     setosa
3          4.7       3.2        1.3       0.2     setosa
4          4.6       3.1        1.5       0.2     setosa
5          5.0       3.6        1.4       0.2     setosa
6          5.4       3.9        1.7       0.4     setosa
7          4.6       3.4        1.4       0.3     setosa
8          5.0       3.4        1.5       0.2     setosa
9          4.4       2.9        1.4       0.2     setosa
10         4.9       3.1        1.5       0.1     setosa
11         5.4       3.7        1.5       0.2     setosa
12         4.8       3.4        1.6       0.2     setosa
13         4.8       3.0        1.4       0.1     setosa
14         4.3       3.0        1.1       0.1     setosa
15         5.8       4.0        1.2       0.2     setosa
16         5.7       4.4        1.5       0.4     setosa
17         5.4       3.9        1.3       0.4     setosa
18         5.1       3.5        1.4       0.3     setosa
19         5.7       3.8        1.7       0.3     setosa
20         5.1       3.8        1.5       0.3     setosa
21         5.4       3.4        1.7       0.2     setosa
22         5.1       3.7        1.5       0.4     setosa
23         4.6       3.6        1.0       0.2     setosa
24         5.1       3.3        1.7       0.5     setosa
25         4.8       3.4        1.9       0.2     setosa
26         5.0       3.0        1.6       0.2     setosa
```

Rename columns with janitor

```
library(janitor)
iris %>% clean_names()
```

```
    sepal_length sepal_width petal_length petal_width    species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

Rename columns with janitor

```
library(janitor)
iris %>% clean_names("lower_camel")
```

```
    sepalLength sepalWidth petalLength petalWidth    species
1           5.1         3.5          1.4         0.2     setosa
2           4.9         3.0          1.4         0.2     setosa
3           4.7         3.2          1.3         0.2     setosa
4           4.6         3.1          1.5         0.2     setosa
5           5.0         3.6          1.4         0.2     setosa
6           5.4         3.9          1.7         0.4     setosa
7           4.6         3.4          1.4         0.3     setosa
8           5.0         3.4          1.5         0.2     setosa
9           4.4         2.9          1.4         0.2     setosa
10          4.9         3.1          1.5         0.1     setosa
11          5.4         3.7          1.5         0.2     setosa
12          4.8         3.4          1.6         0.2     setosa
13          4.8         3.0          1.4         0.1     setosa
14          4.3         3.0          1.1         0.1     setosa
15          5.8         4.0          1.2         0.2     setosa
16          5.7         4.4          1.5         0.4     setosa
17          5.4         3.9          1.3         0.4     setosa
18          5.1         3.5          1.4         0.3     setosa
19          5.7         3.8          1.7         0.3     setosa
20          5.1         3.8          1.5         0.3     setosa
21          5.4         3.4          1.7         0.2     setosa
22          5.1         3.7          1.5         0.4     setosa
23          4.6         3.6          1.0         0.2     setosa
24          5.1         3.3          1.7         0.5     setosa
25          4.8         3.4          1.9         0.2     setosa
26          5.0         3.0          1.6         0.2     setosa
27          5.0         3.4          1.6         0.4     setosa
```

Rename columns with janitor

```
library(janitor)
iris %>% clean_names("title")
```

```
    Sepal Length Sepal Width Petal Length Petal Width    Species
1            5.1         3.5          1.4         0.2     setosa
2            4.9         3.0          1.4         0.2     setosa
3            4.7         3.2          1.3         0.2     setosa
4            4.6         3.1          1.5         0.2     setosa
5            5.0         3.6          1.4         0.2     setosa
6            5.4         3.9          1.7         0.4     setosa
7            4.6         3.4          1.4         0.3     setosa
8            5.0         3.4          1.5         0.2     setosa
9            4.4         2.9          1.4         0.2     setosa
10           4.9         3.1          1.5         0.1     setosa
11           5.4         3.7          1.5         0.2     setosa
12           4.8         3.4          1.6         0.2     setosa
13           4.8         3.0          1.4         0.1     setosa
14           4.3         3.0          1.1         0.1     setosa
15           5.8         4.0          1.2         0.2     setosa
16           5.7         4.4          1.5         0.4     setosa
17           5.4         3.9          1.3         0.4     setosa
18           5.1         3.5          1.4         0.3     setosa
19           5.7         3.8          1.7         0.3     setosa
20           5.1         3.8          1.5         0.3     setosa
21           5.4         3.4          1.7         0.2     setosa
22           5.1         3.7          1.5         0.4     setosa
23           4.6         3.6          1.0         0.2     setosa
24           5.1         3.3          1.7         0.5     setosa
25           4.8         3.4          1.9         0.2     setosa
26           5.0         3.0          1.6         0.2     setosa
27           5.0         3.4          1.6         0.4     setosa
```

What are the advantages/disadvantages of using a function to automatically rename your variables?

Mutate and summarize

- Mutate adds/modifies columns; dataset remains the same size
- Summarize collapses the data set down (by default to 1 row) to calculate summary statistics
- Both functions use a similar form:
  - `mutate(data, new_variable = expression)`
  - `summarize(data, mean = mean(var))`
- Like all other dplyr functions, they return tibbles, so for mutate to "stick" it needs to be assigned back to itself:
  - `data <- mutate(data, var = round(var))`
  - `data <- mutate(data, var_r = round(var))`
- Since summarize collapses to a new level, you might not want to assign it to overwrite the original data set!

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)

ds
```

```
# A tibble: 87 × 4
   name                mass height hair_color
   <chr>              <dbl>  <int> <chr>
 1 Luke Skywalker        77    172 blond
 2 C-3PO                 75    167 <NA>
 3 R2-D2                 32     96 <NA>
 4 Darth Vader          136    202 none
 5 Leia Organa           49    150 brown
 6 Owen Lars            120    178 brown, grey
 7 Beru Whitesun Lars    75    165 brown
 8 R5-D4                 32     97 <NA>
 9 Biggs Darklighter     84    183 black
10 Obi-Wan Kenobi        77    182 auburn, white
# ℹ 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)

ds
```

```
# A tibble: 87 × 5
   in_movie name               mass height hair_color
   <lgl>    <chr>             <dbl>  <int> <chr>
 1 TRUE     Luke Skywalker       77    172 blond
 2 TRUE     C-3PO                75    167 <NA>
 3 TRUE     R2-D2                32     96 <NA>
 4 TRUE     Darth Vader         136    202 none
 5 TRUE     Leia Organa          49    150 brown
 6 TRUE     Owen Lars           120    178 brown, grey
 7 TRUE     Beru Whitesun Lars   75    165 brown
 8 TRUE     R5-D4                32     97 <NA>
 9 TRUE     Biggs Darklighter    84    183 black
10 TRUE     Obi-Wan Kenobi       77    182 auburn, white
# ℹ 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi), .after = "height")

ds
```

```
# A tibble: 87 × 7
   in_movie name               mass height height_m   bmi hair_color
   <lgl>    <chr>             <dbl>  <int>    <dbl> <dbl> <chr>
 1 TRUE     Luke Skywalker       77    172     1.72    26 blond
 2 TRUE     C-3PO                75    167     1.67    27 <NA>
 3 TRUE     R2-D2                32     96     0.96    35 <NA>
 4 TRUE     Darth Vader         136    202     2.02    33 none
 5 TRUE     Leia Organa          49    150     1.5     22 brown
 6 TRUE     Owen Lars           120    178     1.78    38 brown, grey
 7 TRUE     Beru Whitesun Lars   75    165     1.65    28 brown
 8 TRUE     R5-D4                32     97     0.97    34 <NA>
 9 TRUE     Biggs Darklighter    84    183     1.83    25 black
10 TRUE     Obi-Wan Kenobi       77    182     1.82    23 auburn, white
# ℹ 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi), .after = "height") %>%
             arrange(desc(bmi))

ds
```

```
# A tibble: 87 × 7
   in_movie name                  mass height height_m   bmi hair_color
   <lgl>    <chr>                <dbl>  <int>    <dbl> <dbl> <chr>
 1 TRUE     Jabba Desilijic Tiure  1358    175     1.75   443 <NA>
 2 TRUE     Dud Bolt                 45     94     0.94    51 none
 3 TRUE     Yoda                     17     66     0.66    39 white
 4 TRUE     Owen Lars               120    178     1.78    38 brown, grey
 5 TRUE     R2-D2                    32     96     0.96    35 <NA>
 6 TRUE     IG-88                   140    200     2       35 none
 7 TRUE     R5-D4                    32     97     0.97    34 <NA>
 8 TRUE     Jek Tono Porkins        110    180     1.8     34 brown
 9 TRUE     Grievous                159    216     2.16    34 none
10 TRUE     Darth Vader             136    202     2.02    33 none
# ℹ 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi), .after = "height") %>%
             arrange(desc(bmi))

ds <- ds

ds
```

```
# A tibble: 87 × 7
   in_movie name                 mass height height_m   bmi hair_color
   <lgl>    <chr>               <dbl>  <int>    <dbl> <dbl> <chr>
 1 TRUE     Jabba Desilijic Tiure  1358    175     1.75   443 <NA>
 2 TRUE     Dud Bolt                 45     94     0.94    51 none
 3 TRUE     Yoda                     17     66     0.66    39 white
 4 TRUE     Owen Lars               120    178     1.78    38 brown, grey
 5 TRUE     R2-D2                    32     96     0.96    35 <NA>
 6 TRUE     IG-88                   140    200     2       35 none
 7 TRUE     R5-D4                    32     97     0.97    34 <NA>
 8 TRUE     Jek Tono Porkins        110    180     1.8     34 brown
 9 TRUE     Grievous                159    216     2.16    34 none
10 TRUE     Darth Vader             136    202     2.02    33 none
# ℹ 77 more rows
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi), .after = "height") %>%
            arrange(desc(bmi))

ds <- ds %>%
  filter(hair_color %in% c("blond", NA))

ds
```

```
# A tibble: 8 × 7
  in_movie name                  mass height height_m   bmi hair_color
  <lgl>    <chr>                <dbl>  <int>    <dbl> <dbl> <chr>
1 TRUE     Jabba Desilijic Tiure  1358    175     1.75   443 <NA>
2 TRUE     R2-D2                    32     96     0.96    35 <NA>
3 TRUE     R5-D4                    32     97     0.97    34 <NA>
4 TRUE     C-3PO                    75    167     1.67    27 <NA>
5 TRUE     Luke Skywalker           77    172     1.72    26 blond
6 TRUE     Greedo                   74    173     1.73    25 <NA>
7 TRUE     Anakin Skywalker         84    188     1.88    24 blond
8 TRUE     Finis Valorum            NA    170     1.7     NA blond
```

Mutate (add or modify columns)

```
ds <- starwars %>% select(name, mass, height, hair_color)
ds <- ds %>% mutate(in_movie = TRUE, .before = 1)
ds <- ds %>% mutate(height_m = height/100,
                    bmi = mass/(height_m^2),
                    bmi = round(bmi), .after = "height") %>%
             arrange(desc(bmi))

ds <- ds %>%
  filter(hair_color %in% c("blond", NA)) %>%
  mutate(hair_color = ifelse(is.na(hair_color),"no hair", hair_color))

ds
```

```
# A tibble: 8 × 7
  in_movie name                 mass height height_m   bmi hair_color
  <lgl>    <chr>               <dbl>  <int>    <dbl> <dbl> <chr>
1 TRUE     Jabba Desilijic Tiure  1358    175     1.75   443 no hair
2 TRUE     R2-D2                    32     96     0.96    35 no hair
3 TRUE     R5-D4                    32     97     0.97    34 no hair
4 TRUE     C-3PO                    75    167     1.67    27 no hair
5 TRUE     Luke Skywalker           77    172     1.72    26 blond
6 TRUE     Greedo                   74    173     1.73    25 no hair
7 TRUE     Anakin Skywalker         84    188     1.88    24 blond
8 TRUE     Finis Valorum            NA    170     1.7     NA blond
```

Common task: Changing some but not all values within a column

```
ds <- starwars %>% select(name, height, hair_color)

# Alternate: Change all NA values to "no hair", keep all others the same
ds <- ds %>% mutate(hair_color = replace_values(hair_color, NA ~ "no hair"))

# Set all heights greater than 100 to 100, keep all others the same
ds <- ds %>% mutate(height = ifelse(height > 100,100, height))

ds
```

```
# A tibble: 87 × 3
   name               height hair_color
   <chr>               <dbl> <chr>
 1 Luke Skywalker        100 blond
 2 C-3PO                 100 no hair
 3 R2-D2                  96 no hair
 4 Darth Vader           100 none
 5 Leia Organa           100 brown
 6 Owen Lars             100 brown, grey
 7 Beru Whitesun Lars    100 brown
 8 R5-D4                  97 no hair
 9 Biggs Darklighter     100 black
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species, sex)
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species, sex)
ds %>% summarize(min_height = min(height))
```

```
# A tibble: 1 × 1
  min_height
       <int>
1         NA
```

Summarize (calculate variables in aggregate)

```
ds <- starwars %>% select(name, mass, height, species, sex)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))
```

```
# A tibble: 1 × 1
  min_height
       <int>
1         NA

# A tibble: 1 × 1
  min_height
       <int>
1         66
```

Summarize (calculate variables in aggregate)

```r
ds <- starwars %>% select(name, mass, height, species, sex)
ds %>% summarize(min_height = min(height))
ds %>% summarize(min_height = min(height, na.rm = T))

ds %>% summarize(min_height = min(height, na.rm = T),
                 m_height = mean(height, na.rm = T),
                 max_height = max(height, na.rm = T))
```

```
# A tibble: 1 × 1
  min_height
       <int>
1         NA

# A tibble: 1 × 1
  min_height
       <int>
1         66

# A tibble: 1 × 3
  min_height m_height max_height
       <int>    <dbl>      <int>
1         66     175.        264
```

Summarize by group

```
ds %>% group_by(species, sex)
```

```
# A tibble: 87 × 5
# Groups:   species, sex [41]
   name                mass height species sex
   <chr>              <dbl>  <int> <chr>   <chr>
 1 Luke Skywalker        77    172 Human   male
 2 C-3PO                 75    167 Droid   none
 3 R2-D2                 32     96 Droid   none
 4 Darth Vader          136    202 Human   male
 5 Leia Organa           49    150 Human   female
 6 Owen Lars            120    178 Human   male
 7 Beru Whitesun Lars    75    165 Human   female
 8 R5-D4                 32     97 Droid   none
 9 Biggs Darklighter     84    183 Human   male
10 Obi-Wan Kenobi        77    182 Human   male
# ℹ 77 more rows
```

Summarize by group

```
ds %>% group_by(species, sex) %>%
  summarize(min_height = min(height, na.rm = T),
                m_height = mean(height, na.rm = T),
                max_height = max(height, na.rm = T),
                n = n())
```

```
# A tibble: 41 × 6
# Groups:   species [38]
   species   sex    min_height m_height max_height     n
   <chr>     <chr>       <int>    <dbl>      <int> <int>
 1 Aleena    male           79     79           79     1
 2 Besalisk  male          198    198          198     1
 3 Cerean    male          198    198          198     1
 4 Chagrian  male          196    196          196     1
 5 Clawdite  female        168    168          168     1
 6 Droid     none           96    131.         200     6
 7 Dug       male          112    112          112     1
 8 Ewok      male           88     88           88     1
 9 Geonosian male          183    183          183     1
10 Gungan    male          196    209.         224     3
# ℹ 31 more rows
```

Summarize by group

```
ds %>% group_by(species, sex) %>%
  summarize(min_height = min(height, na.rm = T),
                m_height = mean(height, na.rm = T),
                max_height = max(height, na.rm = T),
                n = n()) %>%
  filter(n > 1)
```

```
# A tibble: 8 × 6
# Groups:   species [7]
  species  sex    min_height m_height max_height     n
  <chr>    <chr>       <int>    <dbl>      <int> <int>
1 Droid    none           96     131.        200     6
2 Gungan   male          196     209.        224     3
3 Human    female        150     164.        185     9
4 Human    male          170     182.        202    26
5 Mirialan female        166     168         170     2
6 Wookiee  male          228     231         234     2
7 Zabrak   male          171     173         175     2
8 <NA>     <NA>          157     175         185     4
```

Pay attention to groups/variables following summarize

```
ds %>% group_by(species, sex) %>% summarize(max_height = max(height, na.rm = T))
```

```
# A tibble: 41 × 3
# Groups:   species [38]
   species   sex    max_height
   <chr>     <chr>       <int>
 1 Aleena    male           79
 2 Besalisk  male          198
 3 Cerean    male          198
 4 Chagrian  male          196
 5 Clawdite  female        168
 6 Droid     none          200
 7 Dug       male          112
 8 Ewok      male           88
 9 Geonosian male          183
10 Gungan    male          224
# ℹ 31 more rows
```

Pay attention to groups/variables following summarize

```
ds %>% group_by(species, sex) %>% summarize(max_height = max(height, na.rm = T)) %>% ungrou
```

```
# A tibble: 41 × 3
   species   sex    max_height
   <chr>     <chr>       <int>
 1 Aleena    male           79
 2 Besalisk  male          198
 3 Cerean    male          198
 4 Chagrian  male          196
 5 Clawdite  female        168
 6 Droid     none          200
 7 Dug       male          112
 8 Ewok      male           88
 9 Geonosian male          183
10 Gungan    male          224
# ℹ 31 more rows
```

Pay attention to groups/variables following summarize

```
ds %>% group_by(species, sex) %>% slice_max(height)
```

```
# A tibble: 41 × 5
# Groups:   species, sex [41]
   name                  mass height species   sex
   <chr>                <dbl>  <int> <chr>     <chr>
 1 Ratts Tyerel            15     79 Aleena    male
 2 Dexter Jettster        102    198 Besalisk  male
 3 Ki-Adi-Mundi            82    198 Cerean    male
 4 Mas Amedda              NA    196 Chagrian  male
 5 Zam Wesell              55    168 Clawdite  female
 6 IG-88                  140    200 Droid     none
 7 Sebulba                 40    112 Dug       male
 8 Wicket Systri Warrick   20     88 Ewok      male
 9 Poggle the Lesser       80    183 Geonosian male
10 Roos Tarpals            82    224 Gungan    male
# ℹ 31 more rows
```

What gets tedious...

```
iris %>% clean_names() %>%
  group_by(species) %>%
  summarize(sepal_length_mean = mean(sepal_length),
            sepal_width_mean = mean(sepal_width),
            petal_length_mean = mean(petal_length),
            petal_width_mean = mean(petal_length))
```

```
# A tibble: 3 × 5
  species  sepal_length_mean sepal_width_mean petal_length_mean petal_width_mean
  <fct>                <dbl>            <dbl>             <dbl>            <dbl>
1 setosa                5.01             3.43              1.46             1.46
2 versico…              5.94             2.77              4.26             4.26
3 virgini…              6.59             2.97              5.55             5.55
```

Instead, use across()

```
iris %>% clean_names() %>%
  group_by(species) %>%
  summarize(across(everything(), mean, .names = "{.col}_mean"))
```

```
# A tibble: 3 × 5
  species  sepal_length_mean sepal_width_mean petal_length_mean petal_width_mean
  <fct>                <dbl>            <dbl>              <dbl>            <dbl>
1 setosa                5.01             3.43               1.46            0.246
2 versico…              5.94             2.77               4.26            1.33
3 virgini…              6.59             2.97               5.55            2.03
```

Instead, use across()

```
iris %>% clean_names() %>%
  group_by(species) %>%
  summarize(across(ends_with("length"), list(mean = mean, sd = sd)))
```

```
# A tibble: 3 × 5
  species    sepal_length_mean sepal_length_sd petal_length_mean petal_length_sd
  <fct>                  <dbl>           <dbl>              <dbl>           <dbl>
1 setosa                  5.01           0.352               1.46           0.174
2 versicolor              5.94           0.516               4.26           0.470
3 virginica               6.59           0.636               5.55           0.552
```

Part 3: Code style

Who cares what your code looks like?

- Code that is machine readable might not be human readable
- Human readable code:
  - Helps with code-as-documentation
  - Makes it easier for you to find mistakes
  - Helps with later internal/external code sharing
- Some of my style choices are to make content fit on slides

Spaces

```
# Strive for
z <- (a + b)^2 / d

# Avoid
z<-( a + b ) ^ 2/d

# Strive for
mean(x, na.rm = TRUE)

# Avoid
mean (x ,na.rm=TRUE)
```

Pipes

```
# Strive for
flights %>%
  filter(!is.na(arr_delay), !is.na(tailnum)) %>%
  count(dest)

# Avoid
flights%>%filter(!is.na(arr_delay), !is.na(tailnum))%>%count(dest)
```

Spacing helps group arguments within functions

```
# Strive for
flights %>%
  group_by(tailnum) %>%
  summarize(
    delay = mean(arr_delay, na.rm = TRUE),
    n = n()
  )

# Avoid (lacks visual cue for what is part of summarize)
flights%>%
  group_by(tailnum) %>%
  summarize(
  delay = mean(arr_delay, na.rm = TRUE),
  n = n()
  )
```

Use the styler

- `install.packages("styler")
- Access from addins to style a selection of code or an entire file
- Might need to restart RStudio before using

Data transformation homework assignment

- Use the GitHub repo link below to fork and then clone the project to your own user account and then to work locally on your own computer.
- Be sure you are using R version >= 4.0
- The homework will make sure you learned what we covered, and will also ask you to try out new things to extend your knowledge
- Push your answers to a public Github repo, and turn in the homework by entering the link to the repo on Canvas

Homework Github repo